

Understanding Finance

The Economics of Money

*Two sessions ago you learned to write the numbers.
Last session, to read them. Tonight: what they're worth.*

Every financing conversation you will ever have is someone
pricing your risk.

The banker quoting your loan rate, the investor quoting your valuation, the buyer quoting your exit price — all of them are running the same calculation. Tonight covers that calculation, start to finish.

Tonight's Agenda

I

The Price of Money

Who sells it, who buys it, who bets on it.

II

The Price of Time

A dollar later is worth less than a dollar now — and how much less is the whole game.

III

Pricing the Future

The DCF: how every business gets valued. We build one, live.

IV

Debt vs Equity

The two ways to fund a business — and what each one really costs.

V

How Risk Gets Priced

Why your investor needs 30% when the bank needs 7%.

VI

Exotic Finance Instruments

Options, futures, swaps — and the one idea behind all of them.

ACT I

The Price of Money

Money is a product. It has sellers, buyers — and speculators.

Functions of money.

I

Unit of account

Everything gets one common ruler: dollars.

II

Store of value

Earn today, spend next year. Money carries value through time.

III

Medium of exchange

You trade it instead of bartering bagels for legal work.

IV

Standard of deferred payment

Debts can be written down and settled later — in dollars.

Two of these jobs involve time — carrying value forward, settling debts later. The moment time enters, a question follows: what does it cost to move money through time? That question is tonight's entire session.

Meet the players.



HAS MONEY NOW

Money Sellers

Savers, lenders, bondholders, LPs. They hand money over today and want **more of it back later**. Their question: *what do you pay me to wait — and to risk not getting it back?*



NEEDS MONEY NOW

Money Buyers

Borrowers, founders, home-buyers, CFOs. They need money today and will **pay for the privilege** — in interest, or in a piece of the company. Their question: *what's the cheapest way to rent or raise it?*



WANTS THE OUTCOME

Speculators

Traders and speculators. They don't need the money and aren't lending it — they're **buying and selling the risk itself**, betting on which way a price moves. Their question: *what does the crowd have wrong?*

Which one are you? Trick question — most of you were all three before lunch. A 401(k) makes you a seller. A mortgage makes you a buyer. The March Madness bracket makes you a gambler. Tonight's skill: knowing which seat you're in, and what the person across the table wants.

ACT II

The Time Value of Money

You already know a dollar later is worth less than a dollar now. Tonight we find out how much less.

HANDS UP WHEN YOU'D WAIT

I hand you \$10,000 today —
or you wait one year for...

\$10,200

2% MORE

\$10,500

5% MORE

\$11,000

10% MORE

\$12,000

20% MORE

The number where your hand goes up is your personal price for one year of waiting.

Congratulations — you just quoted a discount rate. No formula required.

Calculating value now and in the future.

Future Value (FV) — what money today becomes later

$$FV = PV \times (1 + r)^n$$

Money moving **forward** through time grows at rate r for n years. \$10,000 today at 10% is worth **\$11,000** in a year — and \$16,105 in five, because each year's growth earns its own growth. That's compounding.

Present Value (PV) — what money promised later is worth today

$$PV = FV \div (1 + r)^n$$

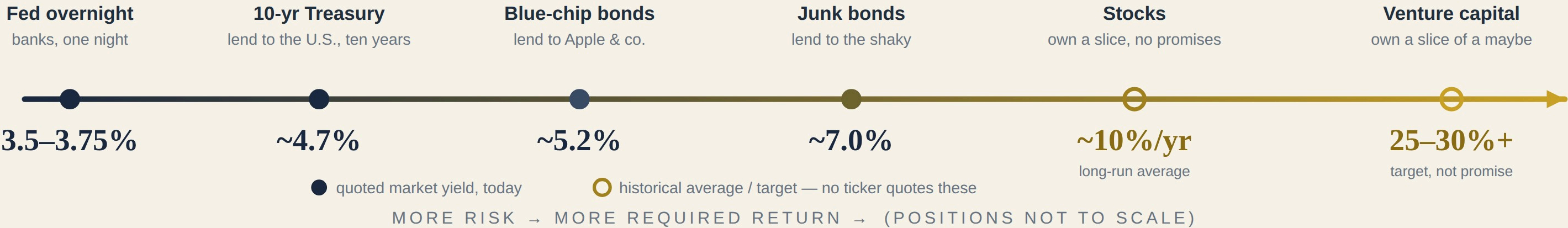
Money moving **backward** shrinks. A promise of \$11,000 next year, at that same 10%, is worth exactly **\$10,000** today. Same formula, run in reverse. Finance calls this *discounting*, and r is the *discount rate*.

Everything tonight runs on these two formulas. Valuing a company, pricing a loan, judging a term sheet — all of it is pulling future dollars back to today. The only argument, ever, is about r : the rate. So where does r come from?

The Spectrum of Returns — every r on one line.

Required return = risk-free rate + $\beta \times$ market risk premium

β ("beta") measures how risky an investment is compared to the market as a whole. The riskier the bet, the higher the β — and the higher the return investors demand. The line below is this equation, drawn left to right.



Every dot answers the same question: **what return do I require to hand my money to someone riskier?** Each step right, the risk rises and the required return rises with it. Keep this line in view — it comes back at the end of the night with more names on it.

ACT III

Pricing the Future

The DCF: how every business on earth gets a price tag. We build one — on a bagel shop you already know.

A DCF answers one question: what is a stream of future cash worth today?

$$NPV = \sum_{t=1}^n CF_t \div (1 + r)^t + TV \div (1 + r)^n$$

DCF — discounted cash flow: each year's cash flow (CF), pulled back to today with the PV formula, then added up. **NPV** — net present value: the total. Three parts, in order:

1

The cash flows

What the business will actually hand you, year by year. You just spent two sessions learning to find this number — it's the cash flow statement, projected forward.

2

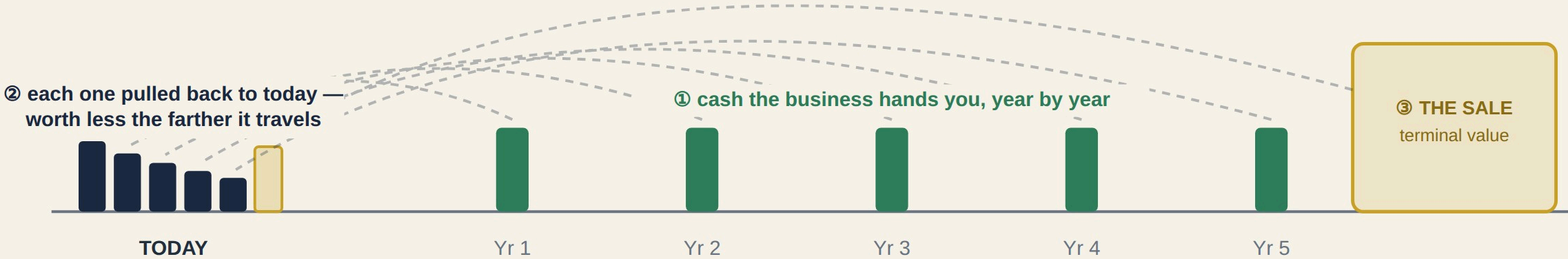
The discount rate

The price of waiting plus the price of risk — picked off the spectrum you just saw. Each year's cash gets pulled back to today with the PV formula — divided by (1+r), once per year of waiting.

3

The terminal value

You can't project forever. At some point you stop — and pretend you sell the whole thing. The sale price at the end is the third part.



The next six slides walk this picture left to right: the flows, the rate, the pull-back, the sale, the answer — then the same math run in reverse.

Franklin's Bagels has a five-year plan.

Franklin's Bagels — projected, \$ thousands

	YR 1	YR 2	YR 3	YR 4	YR 5
Revenue	780	850	1,150	1,650	1,950
locations open	1	1	2	2	2
Free cash flow	95	115	(70)	205	290

WHERE THESE NUMBERS COME FROM

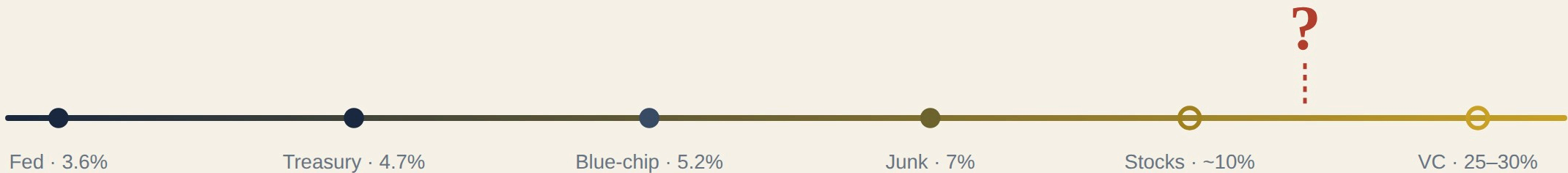
Nobody's filings — **our own projections**. Every line is an assumption. Your job tonight is to attack them: is \$780K plausible for one shop? Does a second location really cost a year of cash? The model is only as honest as the people stress-testing it.

Years 1–2: the one shop you built in Session 04 — already open and running, startup costs behind it. Modest growth, steady cash.

Year 3 — the plot twist: Franklin signs a lease across town. Ovens, buildout, staff hired before revenue: **\$70K of negative cash flow** in the expansion year, even as revenue climbs.

Years 4–5: two shops at full speed. Cash flow triples off the Year 2 base. *If* the plan works.

Impact of risk on money cost.



Supporting Franklin's

- Real cash flow from day one — not a promise of one
- People buy breakfast in every economy
- Assets you can touch: ovens, lease, a known corner

Against Franklin's

- One neighborhood, one owner, one oven fire from zero
- Thin food-service margins, no pricing power
- You can't sell your stake on any screen — illiquid

Call it 20%. Riskier than the whole stock market, not as wild as a venture bet. Notice what just happened: no formula produced that number. **We argued our way to it from the line** — which is how every private-company rate on earth actually gets set. It's an assumption. We'll stress it in the model.

Five years of cash, pulled back to today.

Each year through the machine, at $r = 20\%$

YEAR	CASH FLOW	$\div (1.20)^N$	WORTH TODAY
1	95	$\div 1.20$	79
2	115	$\div 1.44$	80
3	(70)	$\div 1.73$	(41)
4	205	$\div 2.07$	99
5	290	$\div 2.49$	117
Five years of bagels			$\approx \$334K$

Distance shrinks everything. Year 5's \$290K arrives worth \$117K in today's money — the machine took 60% off for five years of waiting and risk.

Even the bad year shrinks. The expansion's \$70K hole costs only \$41K in today's terms — pain that's three years away hurts less than pain today. The machine has no feelings; it discounts losses too.

WHERE THE NUMBERS COME FROM

Cash flows: the Franklin's plan, two slides back. Rate: the 20% this room just argued to. Divisors: $(1.20)^n$ — the reverse gear, n years deep.

At Year 5, we stop projecting — and sell the shop.

Nobody can see Year 12. So the model ends with a pretend sale: hand the business to a buyer at the end of Year 5. What would a buyer pay? In the real world, buyers quote **multiples** — a price expressed as "how many years of X."

MATURE & STEADY → PRICE THE PROFITS

P / E — price to earnings

"I'll pay five years' worth of what it earns." For businesses whose story is the cash they make now.

YOUNG & GROWING → PRICE THE SCALE

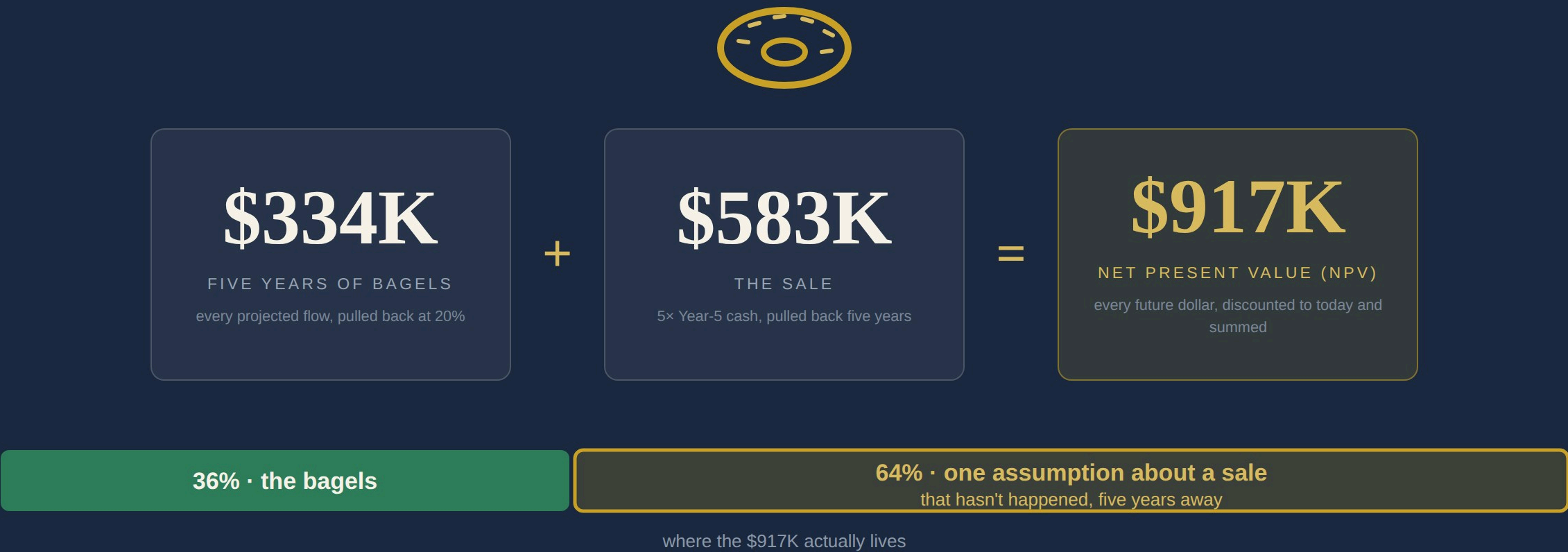
P / S — price to sales

"I'll pay for the revenue and the trajectory." For businesses that are all growth and no profit — yet. (VCs live here.)

$5 \times \$290\text{K Year-5 cash flow} = \$1.45\text{M sale price} \rightarrow \$1.45\text{M} \div (1.20)^5 = \approx \583K today

Calculating the terminal value using a P/E ratio of 5.0 against Year-5 cash flow of \$290,000 yields a sale price of **\$1.45M**. However, that \$1.45M arrives five years from now — so we discount it for time: the present value is **\$583,000**. The 5.0 ratio is our assumption, not a market quote. We test it in the model.

What is Franklin's worth today?



A DCF is mostly one big assumption wearing a lab coat.

That is not a reason to skip the math. It's the reason you **stress** the math — which is exactly what we do next.

Calculating investment return.

IRR — Internal Rate of Return

The annual return a stream of cash flows actually earns on the money invested. Mathematically: the discount rate at which NPV equals exactly zero — the rate where the deal breaks even.

Two questions, one formula

NPV asks: given my rate, what are these cash flows worth? We just answered it — $r = 20\% \rightarrow \$917K$.

IRR asks: given the price, what do these cash flows pay? You pick the dollars; the formula returns a rate.

Franklin asks \$700K for the whole business today $\rightarrow$ his projected stream pays **27.7% per year**

Now the two answers can talk to each other: the stream pays 27.7%, and we demanded 20% — so at \$700K, this deal clears our bar. **This is the language investors speak.** A VC won't say "your NPV is positive at my rate." They'll say "we underwrite to a 30% IRR" — meaning: at the price we're offering, the stream had better pay 30%.

WHERE THE NUMBERS COME FROM

Same five cash flows + the \$1.45M sale, against a \$700K price today. Solve for the rate that balances the scale: 27.7%. (Excel does the solving — you'll see the formula in a minute.)

Let's build a live model of Franklin's Bagels.

Everything you just watched — the flows, the rate, the sale, the pull-back — is sitting in a workbook, wired end to end. Nothing hard-coded. Your assumptions drive every cell.

- **You feed it.** Call out a different rate. A meaner multiple. Push the second shop to Year 4.
- **Watch what moves.** One cell changes; the price tag repaints. Feel which assumptions the value actually hangs on.
- **Claude argues the other side.** We hand the room's assumptions to the AI and ask for the bear case — the buyer trying to pay less.

INTERMISSION

Break — 8 minutes

When we return... digging into the world of financial instruments.

ACT IV

Tools of the CFO — Debt & Equity

Every dollar a business runs on is either rented or sold. The difference decides who owns you.

Understanding leverage.



One house. Two buyers. Five years.

Both buy the same \$500,000 house. Five years later, both sell it for \$700,000.

BUYER A

Writes a check for the whole \$500,000.

No bank, no payments, sleeps like a baby.

BUYER B

Puts \$100,000 down, borrows \$400,000.

30-year mortgage at 6% — \$2,398 a month, every month, for all five years.

VOTE: WHO MADE MORE MONEY?

Lock in your answer before the next slide. Yes, the mortgage payments count.

Everyone who voted is right.

Buyer A — all cash

Cash in, day one	\$500,000
Cash out at sale	\$700,000
Profit	\$200,000
Return on his cash	7.0% / yr

Buyer B — \$100K down + mortgage

Down payment	\$100,000
+ payments, \$28.8K/yr × 5	\$143,900
Total cash in	\$243,900
Sale \$700K – \$372.2K still owed	\$327,800
Profit	\$83,900
Return on his cash	9.2% / yr

A made more dollars. B made more per dollar. "Who made more money" has two honest answers — and finance runs on the second one. B controlled the same \$700K asset with a quarter of the cash, so every point of appreciation landed on a smaller base. That amplification is **leverage**.

IT CUTS BOTH WAYS

Rerun it with the house selling flat at \$500K: A breaks even. B is out \$116,000 — five years of interest, paid for appreciation that never came. Leverage amplifies the direction you're already going. Both directions.

"Wait — what did B do with the other \$400,000?"

Exactly. B didn't burn the cash he didn't put in the house — he still *has* it. Say it goes into the Vanguard Extended Market Index fund the same day, and the market averages 10% a year for the five years (the long-run neighborhood, not a promise).



Leverage didn't create money. It freed B's capital to work two jobs at once — live in the house *and* ride the market. That's the real reason businesses borrow: not because debt is magic, but because the cash it releases can go earn more than the debt costs. When it can't, leverage is just expensive.

Debt is renting money. Equity is selling the machine that makes it.

	Debt	Equity
What it is	A loan. You keep the company.	A sale. They keep a piece — forever.
Duration	Temporary. It has an end date, and then it's gone.	Permanent. No maturity. Your grandkids' company still has these shareholders.
What it costs	A fixed rate — cheaper, because lenders get paid first and can take collateral.	A slice of every future dollar — the most expensive money you'll ever raise, <i>if</i> you succeed.
When things go wrong	Miss payments and they can take the keys. Debt can kill a company.	Nothing is owed. Equity absorbs the bad years — that's what it's paid for.
At the exit	Paid back first, in full, capped there.	Paid last — and gets everything that's left.

Buyer B just showed you why debt is cheaper for the buyer *and* the seller of money. The bank took the safe seat — first in line, house as collateral, 6% and capped there. B kept every dollar of upside past the interest. Equity flips both seats: the investor takes the risk seat, and charges accordingly.

Most companies drink both. Their cost of money is the blend.

$$\text{Cost of capital} = (\text{cost of debt} \times \text{share of debt}) + (\text{cost of equity} \times \text{share of equity})$$

Weight each source's cost by how much of it you use. Finance calls the blend **WACC** — the weighted average cost of capital.

Why it matters

The blend is the hurdle every project inside a company must clear. Money that costs 12% can't fund projects that pay 8%.

Why debt lowers the blend

Debt is the cheaper source — lenders sit in the safer seat, and interest is tax-deductible. Up to the point where the payments can kill you.

Where each cost comes from

Cost of debt is easy to see — it's the interest rate on the loan. Cost of equity is the return investors require for the risk — the β equation from Act II prices it. The full build lands in S07, where the blend becomes the discount rate for your pro forma.

Understanding CFO magic.

$$\text{ROE} = \frac{\text{Net income}}{\text{Revenue}} \times \frac{\text{Revenue}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}$$

The DuPont breakdown of return on equity. Notice Revenue and Assets each cancel out — the three fractions multiply down to Net income ÷ Equity, the return owners actually earn.

Profit margin

Net income ÷ Revenue — how much of each sale you keep. Raise prices, cut costs. **An operating skill.**

Asset turnover

Revenue ÷ Assets — how much revenue each dollar of assets produces. **An operating skill.**

Leverage

Assets ÷ Equity — how much of the asset base is funded with borrowed money. **A financing choice.**

Two of these levers are operating skills — earn more per sale, get more sales from the same assets. The third is a financing choice: run the business on borrowed money. **Leverage multiplies the result of the other two — in both directions.** A spectacular ROE deserves one question first: which lever produced it?

ACT V

How Risk Gets Priced

Why the bank quotes 7% and the venture fund demands 30% — for the same company.

Six criteria impacting your cost of money.

Revenue Quality

Subscription revenue reads safe. Hit-driven, lumpy, one-big-customer revenue reads risky.

Industry Volatility

People buy bagels in recessions. They don't buy yachts. Utilities sit left on the line; crypto exchanges sit far right.

GDP Correlation

If your sales rise and fall with the economy, investors can't hide from you in a downturn — they charge for that.

Debt Levels

Act IV's lesson: leverage amplifies everything — including how risky your equity is to hold.

Liquidity

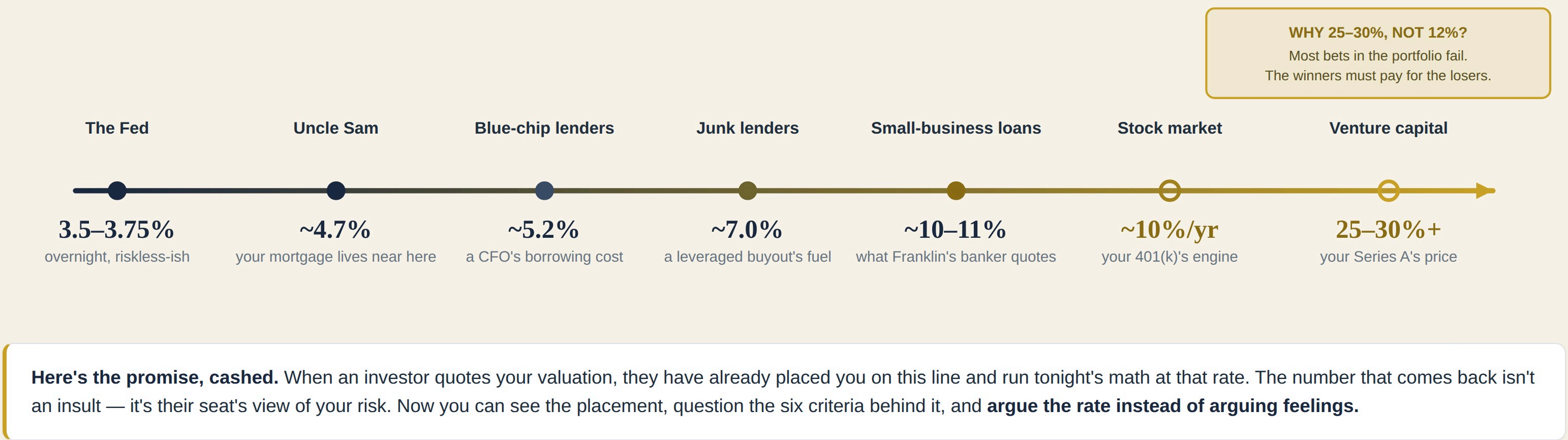
A public stock sells in seconds. A bagel-shop stake sells in months, at a discount, maybe. Illiquidity costs extra.

Exit Date

Cash next quarter vs. cash in Year 9. The further out the payoff, the more heavily it gets discounted.

Wall Street compresses all six into the single number β from Act II. You don't need to compute it — you need the criteria. They're what an investor is silently scoring while you pitch, and every answer moves your rate.

The full spectrum — who charges what, and why.



ACT VI

Exotic Financial Instruments

Ten minutes through the instruments Wall Street builds — and the one idea behind all of them.

You've bought options before. They were just called other things.

A put is an insurance policy

You pay a premium. If disaster hits — the stock craters — you still get to sell at the price the policy guarantees. If nothing happens, the premium is gone and you're glad. **Homeowner's insurance, for a share of stock.**

A call is a deposit that reserves a price

You pay a small fee to lock the right to buy at today's agreed price. If it soars, you buy at the locked price and smile. If it doesn't, you walk away, out only the fee. **Putting a deposit on a house you're not sure about yet.**

Notice what's actually for sale in both: **not the stock — the risk.** The put seller is paid to carry your downside. The call seller is paid to hand over your upside. Once you see risk itself changing hands, every instrument in this act makes sense.

Options in use.

Marcus took his AI-powered SEO company public on NASDAQ in March. He owns a million shares at \$32 — **\$32 million on paper**. The IPO lockup says he can't sell a single share for six months. Every morning he watches a number that is his entire net worth, and he is not allowed to do anything about it.



Buys puts at \$25

THE INSURANCE

whatever happens, he can sell at \$25 — worst case locked at \$25M

paid for by

Selling calls at \$45

THE UPSIDE, SOLD

if it rockets past \$45, the buyer of those calls gets the ride — not Marcus

The call premium pays for the put premium — the trade costs him **nothing out of pocket**. His six months are now bounded: never worse than \$25M, never better than \$45M. **He gave up the upside above \$45 to eliminate the risk below \$25 — at no cost.** The structure is called a collar.

Every instrument, sorted by the risk it prices.

TIME

Bonds, zeros & strips

Pure waiting, sliced by year and sold by the slice. The simplest instrument — you priced one tonight.

CATASTROPHE

Insurance & cat bonds

The hurricane, syndicated. Your premium is a put on your own house; Wall Street scaled the idea to Florida.

CREDIT

Credit default swaps

"Will they pay it back?" — that worry, detached from the loan and sold separately, like insurance on someone else's promise.

RATES & CURRENCY

Swaps

"I'll trade you my floating payments for your fixed ones." Two CFOs trading the risks they'd rather not hold.

PRICE

Futures & options

"What will it cost in June?" — locked in today. Flour for Franklin, fuel for airlines, Marcus's collar.

BELIEF

Prediction markets

The election, priced by the minute. Pure disagreement, wearing a ticker.

Every one of these is somebody's risk, carved out and sold to somebody willing to hold it — priced with tonight's math.

The model leaves with you tonight.

The Franklin's workbook — assumptions, P&L, cash flow, valuation, every formula annotated — comes home with you. Three experiments before we meet again:

1

Move r

Run 15% and 25%. Watch what a five-point argument about risk does to the price tag — that's every valuation negotiation you'll ever have, in one cell.

2

Delay the second shop to Year 4

Does the value drop as much as you'd guess? Timing is a lever too.

3

Cut the exit to 3×

If the terminal multiple falls, is Franklin's still worth buying at \$700K? Now you're stress-testing the model's biggest assumption.

Bring your results to S07 — next session you build the pro forma that feeds this model, line by line.

Money has a price.
Time has a price.
Risk has a price.

Every loan offer, term sheet, and acquisition price is built with the math you saw tonight — someone picks a rate, discounts your future, and hands you the result. Now you can check their work, from either side of the table.